

TITLE PAGE

Title: **Hyperalgesia and Neuropathic Pain Mechanism in Rodent Models of Cigarette Smoking- and Nicotine-Induced Precipitated Withdrawal Study**

Authors: Vishakha Shrimali^{1,2}, Deepsi Rathore^{1,2}, Aanchal Joshi¹, Nibedita Naha^{1,✉}

Affiliation: ¹Biochemistry Dept., Biological Sciences Div., ICMR-National Institute of Occupational Health (NIOH), Ahmedabad 380016, Gujarat, INDIA. ²Life Sciences Dept., Gujarat University, Ahmedabad 380009, Gujarat, INDIA.

Short Title: Hyperalgesia, neuropathic pain in nicotine withdrawal, smoking cessation.

✉ **Correspondence:** Dr. Nibedita Naha, Scientist E and Head, Biochemistry Dept., Biological Sciences Div., ICMR-National Institute of Occupational Health (NIOH), Ahmedabad 380016, Gujarat, INDIA. Phone: +91-79-22688737. Email: nibedita.72@yahoo.com; naha.n@gov.in. URL: <https://nioh.org/>

Orchid ID: VS, 0000-0003-0778-0209. **DR**, 0000-0001-7467-5455. **AJ**, 0009-0009-1012-4510. **NN**, 0000-0001-8239-1655.

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ABSTRACT PAGE:

Hyperalgesia and Neuropathic Pain Mechanism in Rodent Models of Cigarette Smoking- and Nicotine-Induced Precipitated Withdrawal Study

Abstract: 228 words

Introduction: Quitting smoking, or nicotine deprivation, is associated with withdrawal symptoms, including increased pain sensitivity and hyperalgesia through poorly managed current therapies, thus highlighting the need to improve our knowledge of pain mechanisms and its management during nicotine withdrawal, or smoking cessation states. **Aims and Methods:** In this context, the present study investigates the role of brain-derived neurotrophic factor (BDNF), inflammation, and redox balance in the cerebral cortex of adult rats exposed to oral nicotine and passive cigarette smoking through a whole-body inhalation chamber for short- and long-term exposure, where doses closely mimic the human smoking scenario. A subset of the exposure group received a single dose of mecamylamine hydrochloride, a nicotine acetylcholine receptor blocker, to evaluate the precipitated withdrawal effects as contributors to hyperalgesia and neuropathic pain. **Results:** The results reveal that nicotine/cigarette smoking induces hyperalgesia, lowers pain tolerance, and upregulates BDNF expression in adult rats as compared to the unexposed healthy controls. Mecamylamine mitigates the negative effects of nicotine by modulating neural circuits of the cerebral cortex involved in pain and inflammation. Specifically, mecamylamine changes cytokine dynamics by upregulating interleukin-6 (IL-6) expression, restoring redox balance, and downregulating BDNF and its receptor, Trk- β , in the cerebral cortex, thereby exacerbating pain sensitivity and ameliorating pain tolerance. **Conclusion:** The BDNF-Trk- β /IL-6 cascade in the cerebral cortex may prevent nicotine relapse by alleviating hyperalgesia and neuropathic pain during smoking cessation/nicotine withdrawal.

Implications: 80 words

Tobacco use is almost double in chronic pain patients, where opioid-based drugs cause more addiction burden. The study provides an insight into potential non-opioid therapeutic strategies targeting the BDNF-Trk- β /IL-6 cascade in the cerebral cortex, the higher center of behavioral activities, using adult rodent models where nicotine doses closely mimic the human smoking situation, including passive cigarette smoking and oral nicotine use. The data of the study may be useful for neuropathic pain management during nicotine withdrawal/smoking cessation and quitting efforts.

Keywords: 8 words

Nicotine withdrawal; Smoking cessation; Mecamylamine; Hyperalgesia; Neuropathic pain; BDNF-Trk- β -IL-6 cascade; iNOS-Nrf2 levels; Cerebral cortex.

Abbreviations: 17 abbreviations

ANOVA, One-way analysis of variance; BDNF, Brain derived neurotrophic factor; BW, Body weight; CCSEA, Committee for the Control and Supervision of Experiments on Animals; CNS, Central nervous system; i.p., Intraperitoneal injection; IFN- α , Interferon-alfa; IHC, Immunohistochemistry; IL, Interleukin; iNOS, inducible nitric oxide synthase; nAChR, Nicotinic acetylcholine receptors; nNOS, Neural nitric oxide synthase; Nrf2, Nuclear factor-erythroid 2-related factor 2; PBS, Phosphate buffered saline; PNS, Peripheral nervous system; SEM, Standard error of mean; Trk- β , Tyrosine kinase-beta.

Highlights: 112 words

1. Quitting smoking, or nicotine deprivation associated to withdrawal symptoms like, increase pain sensitivity and hyperalgesia, though poorly managed by current therapies.
2. Passive cigarette smoking and oral nicotine exposure trigger nAChR, causing hyperalgesia and neuropathic pain in adult rats, where doses closely mimic the human smoking situation.
3. Single post-exposure dose of mecamylamine (i.p.) improves thermal pain sensation by regulating IL-6 and BDNF-Trk- β signalling.

4. BDNF-Trk- β and IL-6 cascade could be a non-opioid therapeutic target for cigarette smoke/nicotine-induced hyperalgesia and neuropathic pain.
5. The study throws a light on pain management by preventing nicotine relapse; thus may have a possibility to enhance the success rates of quitting, or smoking cessation efforts.

Perspective: 63 words

BDNF-Trk- β -IL-6 cascade may have a therapeutic potential targeting to reduces neuropathic pain, alleviates hyperalgesia and prevents nicotine relapse, so far as the smoking cessation and nicotine withdrawal is concerned. Further, the present research lays the groundwork for creating a focused non-opioid intervention strategy addressing both the aspects of nicotine addiction and pain management in order to exacerbate smoking cessation, or quitting success rates.

Hyperalgesia and Neuropathic Pain Mechanism in Rodent Models of Cigarette Smoking- and Nicotine-Induced Precipitated Withdrawal Study**Introduction**

Psychostimulants, nicotine/smoking significantly influence pain sensitivity, contributing to chronic pain and hypersensitivity.¹ USDHHS classifies nicotine and/or smoking as highly addictive substances,² indicating that heightened pain sensitivity is common in addiction, tolerance, and dependence.³ Tobacco addiction and neuropathic pain frequently coexist, creating a self-reinforcing cycle where chronic pain exacerbates tobacco use and vice versa, resulting in increased suffering, dependency, and decreased quality of life.^{4,5} Smokers experience a higher prevalence of chronic pain, characterized by more severe pain and a greater number of painful sites than the non-smokers.^{1,6-8} Long-term exposure to nicotine, an agonist of nicotinic acetylcholine receptors (nAChR), interferes with pain processing and contributes to chronic pain during prolonged use;^{9,10} also exacerbates pain perception during withdrawal.¹¹ Chronic pain is intensified in initial phases of nicotine withdrawal; however, regardless of duration of nicotine abuse, smokers are vulnerable to hyperalgesia at the early stages of nicotine withdrawal, affecting neural pain pathways.¹¹ Further, tobacco use is almost double in chronic pain patients where opioid-based drugs cause more addiction burden.^{12,6} Though the precise mechanism linking nicotine-induced hyperalgesia to chronic/neuropathic pain remains unclear, underscoring the need for identifying novel molecular targets to address the situation during smoking cessation/nicotine withdrawal conditions.

Nicotine/cigarette smoking elevates brain-derived neurotrophic factor (BDNF) expression while downregulating nuclear factor erythroid 2-related factor 2 (Nrf2) in the cerebral cortex of adult rats.¹³ BDNF overexpression is associated with increased pain perception, mediated

through its receptor tyrosine kinase-beta (Trk- β).^{14,15} Nicotine and other psychostimulants increase nAChR expression, contributing to pain, addiction, tolerance, and withdrawal behaviors.¹⁶ Chronic neuroinflammation, driven by elevated BDNF in the central and peripheral nervous systems (CNS, PNS), plays a critical role in neuropathic pain sensitivity.¹⁷⁻¹⁹ Tobacco exposure remarkably increases the pro-inflammatory cytokine, interleukin-6 (IL-6), while decreasing during withdrawal.²⁰ Additionally, IL-6 directly induces neuropathic pain by its nociceptive action.²¹ BDNF and IL-6 are considered as primary biomarkers for neuropathic pain, whereas Trk- β , agonist of BDNF, is an indirect pain biomarker.^{15,22} Further, nitric oxide is an important molecule in neuropathic pain. Neural and inducible isoforms of nitric oxide synthase (nNOS, iNOS)-positive neurones are found in sciatic nerve inflammation/injury.²³ Pro- and anti-inflammatory cytokine balance during nicotine-induced neuroinflammation is driven by free radicals, including iNOS, resulting in inhibition of signaling pathways.^{24,25} Though significant pain reduction with concomitant improvement of neuronal function is mediated through Nrf2 activation,²⁶ further research is necessary to find out the redox balancing role of iNOS-Nrf2 in brain cells during neuropathic pain and its clinical utility. Also, chronic, or neuropathic pain during nicotine withdrawal is a significant barrier to smoking cessation, making it challenging for individuals to quit.^{6,12}

Thus, understanding the mechanisms during hyperalgesia and neuropathic pain associated with nicotine withdrawal is a critical medical priority and strengthens the need for the present study so far as the non-opioid intervention is concerned to overcome the addiction burden of the opioid-based therapy.^{6,12} Therefore, a specific blocker of nAChR, mecamylamine hydrochloride, is used in the present study for novel non-opioid therapeutic intervention to address the nicotine-induced precipitated withdrawal and underlying mechanism. In the present study, mecamylamine significantly blocks nAChR and effectively mitigates nicotine-/cigarette smoke-induced pain sensitivity and inflammatory responses in the preclinical adult rodent

models that closely mimic real-life human smoking situations. The findings highlight the BDNF-Trk- β /IL-6 cascade as a promising non-opioid therapeutic target for hyperalgesia and neuropathic pain in the context of smoking cessation/nicotine withdrawal.

Materials and Methods

Chemicals and drugs: Chemicals, reagents, buffers, and drugs such as (-)-nicotine $\geq 99\%$ (GC), liquid, CAS No. 54-11-5 and mecamylamine hydrochloride $\geq 98\%$ (TLC), powder, CAS No. 826-39-1, were procured from Sigma-Aldrich (St. Louis, MO, USA), and ThermoFisher Scientific (Waltham, MA, USA), unless otherwise specified. Primary antibodies were obtained from Santa Cruz Biotechnology (Dallas, TX, USA) and species specificity was checked as per the datasheet provided. All buffers used in the study were supplemented with protease and phosphatase inhibitor cocktails from Sigma-Aldrich (St. Louis, MO, USA). Drugs were dissolved in sterile phosphate-buffered saline (PBS).

Study design: Albino Wistar rats of 250–300 g body weight (BW) were housed in the ICMR-NIOH animal facility under standard conditions (12-hr dark/light cycles, $23 \pm 3^\circ\text{C}$ temperature, $55 \pm 15\%$ relative humidity) with food and water *ad libitum* as mentioned earlier.¹³ After a five-day acclimatization period, the rats were randomly allocated into two main groups: - a control group (n=10) and an experimental group (n=90) for 4 and 12 weeks' study. The experimental group was further divided into two subgroups: - a smoking group (n=30) and a nicotine group (n=60). Rats in the smoking group were exposed to passive inhalation of two commercially available filtered cigarettes (ITC, India) for 1 hr per day in a tightly sealed whole-body inhalation chamber, following the method described in a previous study.¹³ In the nicotine group, rats received nicotine at doses of 6.0 and 9.0 mg/kg BW/day via gavage after dissolving it in sterile PBS, as per established protocol.¹³ At the end of the exposure period, the cigarette smoking (n=20) and nicotine-exposed (n=40) groups were treated with the nAChR blocker

mecamylamine hydrochloride, at doses of 1.5 and 3.0 mg/kg BW via intraperitoneal injection (i.p.) as per the precipitated withdrawal study regimen, based on the previous studies.²⁷⁻²⁹ In the present study, withdrawal is induced by a single dose of mecamylamine at the end of the cigarette smoking/nicotine exposure for 4 and 12 weeks to address the novel non-opioid intervention strategy during smoking cessation, for which dependence was already developed.³⁰ After 20 min., each individual rat was evaluated for hyperalgesia. The study design and distribution of animals across each group were mentioned in **Table 1** and **Fig. S1**. The nicotine dosages in this study were closely mimicking the amount of nicotine intake as with the cigarette smoke in humans;³¹ hence, the rodent model used here may closely resemble the human smoking situation.¹³ Further, humans are also exposed to chewing tobacco form, where the route of nicotine entry is oral; therefore, in the present study, both the oral and smoking forms of nicotine's effects were compared in the above perspective. Following the hyperalgesia assessment, the rats were euthanized using ketamine and xylazine, and the cerebral cortex was collected aseptically, as described in a previous study.¹³

Ethics Statement: The study/dosing was performed in accordance with the statement of ethical standards of the Committee for the Control and Supervision of Experiments on Animals (CCSEA), Ministry of Fisheries, Animal Husbandry and Dairying, Dept. of Animal Husbandry and Dairying, Govt. of India and ARRIVE 2.0 guidelines. Throughout the study, the blinding policy of data collection and analysis was followed. The carcasses were disposed of by a private biomedical waste disposal agency as per the local municipal regulations (Govt. of Gujarat, India) and approved by the host institute (Govt. of India).

Hyperalgesia: Hyperalgesia was assessed by a thermal pain model: - tail-flick test using hot water of 55°C, and the latency of the avoidance response to thermal stimulus (nociceptive response) was recorded before and after nAChR blocker, mecamylamine treatment using a stop watch.³⁰ Briefly, the individual rat was held steadily by the back in one hand, and the tip of the

tail was dipped into the hot water until it reacted by flicking the tail and/or trying to jump due to the feeling of discomfort, i.e., measured the withdrawal latency time in sec. To avoid errors, the hyperalgesia study was done by the same authorized person for all the animals.

Immunohistochemistry (IHC): To elucidate the underlying mechanism of hyperalgesia and neuropathic pain during precipitated withdrawal in the cigarette smoking- and nicotine-induced pre-clinical adult rodent models, a double IHC technique was performed on paraffin-embedded tissue sections, as described previously.^{13,32} Briefly, after deparaffinization by conventional process, antigen retrieval was done on the sections using sodium citrate buffer (10 mM, pH 6.0) at 90°C for 10 min. After cooling to room temperature, the slides were treated with hydrogen peroxide dissolved in methanol to quench the endogenous peroxides, followed by washing with the sterile PBS and blocking with the horse serum according to the VECTASTAIN Universal IHC kit protocol (Vector Laboratories, Burlingame, CA, USA). Mouse- and rabbit-derived anti-BDNF (N-20), anti-Trk- β (F1), anti-nAChR(G10), and anti-IL-6 (C12-1-hIL-6) antibodies were used for this purpose in 1:50 dilution, followed by treatment with cobalt-labelled universal secondary antibody (1:100 dilutions; Vector Laboratories) as per standard protocol.¹³ Brain sections from the healthy unexposed control rats treated with only the secondary antibody served as negative controls (**Fig. S2**). The stained slides were examined under a bright-field microscope, and images of the same magnifications among different groups were captured by an RTC-7 microscope attached with a digital camera (Radical Scientific, India). For quantitative analysis of colocalization of the signaling molecules in the cerebral cortex, 10 randomly selected areas (1.5×1.5 cm each) from different groups were analyzed as described in our earlier work.³² Immunoreactivity was quantified based on the expression of target proteins, indicated by brown/black staining in the cerebral cortex.

Redox homeostasis: To investigate whether oxidative stress, or precisely, the redox balancing role of iNOS and Nrf2 influences hyperalgesia and neuropathic pain in the cigarette smoking-

and nicotine-induced pre-clinical precipitated withdrawal models, the expressions of iNOS and Nrf2 (anti-iNOS (C-11) and anti-Nrf-2 antibodies; 1:50 dilutions) were assessed using double IHC on cerebral cortex tissue sections. Immunoreactivity was quantified as described previously.^{13,33}

Statistical analysis: One-way analysis of variance (ANOVA) followed by Tukey's HSD *post hoc* test for multiple comparisons between different groups were performed using VassarStats.Net software (<http://vassarstats.net>). Results were expressed as mean±standard error of mean (SEM) of the three independent experiments, as we did earlier.^{13,32} The total number of animals in each group (n) was specified in the legends. Although data distribution was assumed to be normal, this assumption was not formally tested. Hence, to confirm whether the data was skewed, or not, a Mann-Whitney U test was conducted using SPSS, ver. 26.0, and the results were expressed as U- and Z-scores. A '*p*' value of <0.05 was considered statistically significant and presented alongside the data.^{13,32,33}

Results

Effect of dosing on food and water consumption, and BW of animals: There were no significant differences in average BW gain, food intake, water consumption, or individual organ weights between the experimental and control groups in 4- and 12-weeks' treatment protocols (*data not shown*). These findings are consistent with our previous studies.^{13,32,33} Also, the rats remained healthy throughout the study period, and no abnormalities of physical appearance, urination, defecation, locomotion, and grooming behaviors were observed.

Mecamylamine improves thermal pain sensitivity: To address the thermal hyperalgesia during precipitated withdrawal in the passive cigarette smoking- and oral nicotine-induced pre-clinical adult rodent models, a tail-flick test was performed, and latency of avoidance to thermal stimulus was recorded. **Fig. 1** shows a significant increase in latency to thermal pain response

in the cigarette smoke-exposed groups as compared to the healthy unexposed controls. However, a single i.p. injection of the nAChR blocker, mecamylamine, at two different doses, i.e., 1.5 and 3.0 mg/kg BW, at the end of the exposure period significantly revived the sensitivity to thermal pain as compared to the cigarette smoking groups and control groups in 4 weeks [$F(3,16)=12.97, p<0.0001$ vs. *Crt, CS, CS Me1.5, CSMe3.0*; **Fig. 1A**] and 12 weeks [$F(3,16)=70.97, p<0.0001$ vs. *Crt, CS, CS Me1.5, CSMe3.0*; **Fig. 1B**]. The similar trend of remarkably higher sensitivity towards thermal pain was observed in the oral nicotine exposed groups at doses of 6.0 and 9.0 mg/kg BW/day for a period of 4 weeks and 12 weeks as compared to the healthy unexposed control rats. This increased sensitivity was found to be reduced after mecamylamine treatment in both the nicotine doses of 4 weeks [$F(3,16)=10.82, p<0.0003$ vs. *Crt, N6, N6 Me1.5, N6 Me3.0*; $F(3,16)=19.8, p<0.0001$ vs. *Crt, N9, N9 Me1.5, N9 Me3.0*; **Fig. 1C**] and 12 weeks [$F(3,16)=14.88, p<0.0003$ vs. *Crt, N6, N6 Me1.5, N6 Me3.0*; $F(3,16)=15.36, p<0.0001$ vs. *Crt, N9, N9 Me1.5, N9 Me3.0*; **Fig. 1D**]. Further, the above significant data of thermal hyperalgesia (ANOVA) of the adult rats obtained from the tail-flick test was also found to be significant by the Mann-Whitney U test (**Table S1**).

Mecamylamine influences neuropathic pain biomarkers: To study the expression scenario of neuropathic pain biomarkers in cigarette smoking- and oral nicotine-exposed pre-clinical adult rodent models of precipitated withdrawal study with respect to thermal hyperalgesia as discussed above, a double IHC was done. For this purpose, colocalization of neuropathic pain biomarkers such as BDNF and IL-6,^{15,21} and an indirect pain biomarker, Trk- β ,²² in the cerebral cortex under different experimental conditions were studied. Only significant and representative groups based on the hyperalgesia study were used in the IHC analysis to quantify protein expressions in the cerebral cortex and to examine their expression levels in pain sensitivity. The cerebral cortex offers a more comprehensive and integrative environment for evaluating neuroadaptation related to pain perception and effective processing during nicotine

withdrawal-domains,¹⁹ beyond the specialized roles of the nucleus accumbens, locus coeruleus and prefrontal cortex, as these areas are more integrated for addiction, attention or arousal, and higher-level cognitive functions.³⁴⁻³⁶ **Fig. 2A** demonstrates that the expression of nAChR was significantly upregulated in cigarette smoking and oral nicotine-exposed groups of 4 and 12 weeks as compared to the respective controls, which was correlated with BDNF expressions in both the cigarette smoking- and nicotine exposed groups, irrespective of the duration of exposures. Further, a similar correlation was observed in the mecamylamine-treated groups upon cigarette smoking and oral nicotine exposure (9.0 mg/kg BW). However, as shown in **Fig. 2B**, the expressions of nAChR and BDNF were significantly downregulated by a single i.p. injection of mecamylamine as compared to their respective cigarette smoke exposure groups of 4 weeks [$F(4,45)=7.56$, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, *nAChR*; $F(4,45)=9.75$, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, *BDNF*; **Fig. 2B**] and 12 weeks [$F(4,45)=4.6$, $p<0.0033$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, *nAChR*; $F(4,45)=5.11$, $p<0.00176$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, *BDNF*; **Fig. 2B**]. Moreover, 9.0 mg/kg BW/day nicotine exposure revealed a significant reduction of nAChR and BDNF expressions upon mecamylamine treatment only in 4 weeks [$F(4,45)=7.56$, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, *nAChR*; $F(4,45)=9.75$, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, *BDNF*; **Fig. 2B**] as compared to the exposure alone, which may be due to the development of tolerance with increasing duration of exposures, i.e., 12 weeks.³⁷

BDNF is the specific agonist for the Trk- β receptor, due to which its expression levels are in accordance with the Trk- β protein expression.¹⁴ As expected, **Fig. 2C** represents the similar trend of expression pattern of BDNF and Trk- β in the cigarette smoking- and nicotine-exposed groups, independent of the duration of exposure. Both these proteins were significantly upregulated in the cigarette smoking and nicotine-exposed (9.0 mg/kg BW) groups as compared to the healthy unexposed control groups, which were significantly downregulated

upon single dose of mecamlamine treatment in the respective subgroups of 4 weeks [F(4,45)=12.54, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, BDNF; F(4,45)=4.38, $p<0.0044$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, Trk- β ; **Fig. 2D**] and 12 weeks [F(4,45)=5.75, $p<0.0007$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, BDNF; F(4,45)=4.1, $p<0.0064$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, Trk- β ; **Fig. 2D**]. Elevated BDNF levels can contribute to neuroinflammation and may be responsible for neuropathic pain.¹⁵ The expression of IL-6, an anti-inflammatory and indirect pain marker, was elevated in cigarette smoking and oral nicotine exposure groups (9.0 mg/kg BW) as compared to the healthy unexposed controls of 4 weeks (**Fig. 2E**). This elevation was found to be significant in the 12 weeks' exposure group only when compared with the unexposed controls. IL-6 was downregulated with concomitant reduction of BDNF expression by treatment with mecamlamine, an nAChR blocker, in both the cigarette smoking and nicotine groups upon 4 weeks [F(4,45)=17.69, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, IL-6; F(4,45)=13.61, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, BDNF] and 12 weeks [F(4,45)=13.32, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, IL-6; F(4,45)=7.95, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*, BDNF] study (**Fig. 2F**). The significant coexpressions of all the above proteins of interest (ANOVA) in the cerebral cortex of the adult rats were also cross-checked by the Mann-Whitney U test (**Table S2**).

Mecamlamine conserves key markers for redox homeostasis: The analysis of Nrf2 activity can provide information about the cellular response to oxidative stress, as it can detect changes during its activation, or nuclear translocation, which may affect the regulation of its upstream genes involved in cellular defence mechanisms against oxidative stress caused by increased activity of iNOS.^{13,32} To find out the redox balancing potential of Nrf2 and iNOS during neuropathic pain of the precipitated withdrawal study upon cigarette smoke/nicotine exposure, a double IHC is performed, and colocalization of both the proteins in the cerebral cortex of the adult rats was studied by assessing their immunoreactivity. **Fig. 3A** reveals iNOS

and Nrf2 immunoreactivity in the cerebral cortex of the adult rats during cigarette smoking and nicotine exposure, either alone; or with nAChR blocker, mecamylamine post-exposure treatment. Cigarette smoking and nicotine exposure (9.0 mg/kg BW) cause significant overexpression of iNOS in the cerebral cortex as compared with the respective controls of 4 weeks [$F(4,45)=14.04$, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*] and 12 weeks [$F(4,45)=18.08$, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*] groups (**Fig. 3B**). Nrf2, the master cyclor of cellular antioxidants, was remarkably expressed in the mecamylamine-treated subgroups as compared to the respective exposure groups in 4 weeks [$F(4,45)=10.63$, $p<0.0001$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*] and 12 weeks [$F(4,45)=4.68$, $p<0.00303$ vs. *Crt*, *CS*, *CS Me3.0*, *N9*, *N9 Me3.0*] study period (**Fig. 3B**). Finally, the significant coexpressions of iNOS and Nrf-2 (ANOVA) in the cerebral cortex of the adult rats were verified by the Mann-Whitney U test (**Table S3**).

Discussion

Smoking cessation is a major threat to public health, often intricate by the onset, or exaggeration of neuropathic pain, thereby rendering smokers more susceptible to pain.³⁸ In order to mitigate withdrawal-induced hyperalgesia and improve quitting results it may be essential to target neuroinflammatory pathways as well as neuropathic pain biomarkers due to their link with nicotine craving and relapse during withdrawal.^{15,21,39} With an emphasis on the complex interactions between oxidative stress and neuroinflammation in neuropathic pain processing, the present work addresses the precise molecular mechanisms behind nicotine withdrawal-induced hyperalgesia in the cerebral cortex of the adult rats with respect to nAChR and BDNF-Trk- β signaling because of their relation with the anti-inflammatory cytokine, IL-6. According to earlier studies, smokers perceive pain more intensely when exposed to nicotine than the healthy non-smokers.^{38,40} Moreover, nociceptive responses in a variety of pain

assessment tests, such as hot plate, tail-flick, etc., are consistently improved by nicotine withdrawal, as observed in several pre-clinical animal models.^{41,42}

In the present study, the tail flick method is considered to assess the thermal hyperalgesia developed during precipitated withdrawal in the passive cigarette smoking- and oral nicotine-induced pre-clinical adult rodent models. In line with the previous research,¹⁰ present findings showed that irrespective of the duration of exposure, cigarette smoking and nicotine (9.0 mg/kg BW) have low tolerance for pain sensitivity, as exhibit less withdrawal latency time compared to the healthy unexposed controls (**Fig. 1A-D**). nAChR, being a specific agonist for nicotine, are thought to be a promising target for developing non-opioid therapeutic treatment for neuropathic pain.^{10,43} Further, single post-exposure i.p. administration of mecamylamine, increases tail withdrawal latency of the adult rats at 1.5 and 3.0 mg/kg BW doses (**Fig. 1A-D**), resulting in reduction of thermal hyperalgesia upon precipitated withdrawal. This implies that mecamylamine, a non-competitive nAChR antagonist, may interfere with downstream processes of increased pain sensitivity, underscoring its potential as a non-opioid treatment option for nicotine dependence and withdrawal-associated neuropathic pain. Though human pain perception is elevated when nAChR is absent,⁴³ research indicates optimal nicotine levels may reduce the pain sensitivity.¹⁰ However, this seeming paradox requires more detailed study with a large number of animals addressing the intricate connection between nicotine, nAChR, and pain regulation during nicotine addiction-tolerance-withdrawal situations.

Psychostimulants like nicotine, cocaine, or morphine increase nAChR expression in the cerebral cortex.¹⁶ In the present study, the expression of nAChR through double IHC techniques is assessed in the cerebral cortex of the adult rat brain, which is upregulated in both cigarette smoking and nicotine exposures (9.0 mg/kg BW) as compared to the healthy unexposed controls (**Fig. 2A-B**). This event is counterbalanced by a single post-exposure treatment of mecamylamine (3.0 mg/kg BW), irrespective of the duration of exposures. However, at high

nicotine exposure (9.0 mg/kg BW, 12 weeks), no change of nAChR expression is observed (**Fig. 2B**), suggesting over time higher concentrations of nicotine in the brain tissue cause desensitization and downregulation of the nAChR towards the responsiveness of nicotine, while necessitating increasing the intake.³⁷

Moreover, high nAChR expression directly influences the BDNF expression in the cerebral cortex,⁴⁴ though to date no study has been done to elucidate the co-localization of nAChR, BDNF, and Trk- β expressions in the cerebral cortex.¹⁴ BDNF is a key modulator of both inflammatory and neuropathic pain,^{19,22} as enhanced BDNF expression is responsible for greater pain perception in the exposed models.¹⁵ Our results are in corroboration with the previous findings, where BDNF expression is increased during cigarette smoke and nicotine exposure (9.0 mg/kg BW) as compared with the control group, which is reversed upon post-exposure single-dose mecamylamine (3.0 mg/kg BW) treatment in both the 4- and 12-weeks' studies (**Fig. 2B**). Also, this is the first-ever study that shows co-localization pattern of nAChR with BDNF among the healthy unexposed controls, cigarette smoke/nicotine-exposed and post-exposure single-dose mecamylamine-treated groups (**Fig. 2A-B**). Further, at high nicotine exposure (9.0 mg/kg BW, 12 weeks), no change was found in the expression of the BDNF, as its expression is regulated by the nAChR.⁴⁴ BDNF binds with and activates its specific selective receptor, Trk- β , resulting in modulation of its own cellular expression. Results obtained from the present study show that the Trk- β expression is increased by high BDNF expression during cigarette smoking and nicotine exposure (9.0 mg/kg BW) as compared with the unexposed healthy controls, which is reversed by mecamylamine (3.0 mg/kg BW) in both 4 and 12 weeks (**Fig. 2C-D**). This is further supported by the upregulation of Trk- β expression,¹⁴ indicating that BDNF overflow in the above nicotine- exposed group was carried out via the Trk- β receptor only. In contrast, single post-exposure dose of mecamylamine (3.0 mg/kg BW) is sufficient to lower both the nAChR and Trk- β expressions during cigarette smoking and

nicotine exposure (9.0 mg/kg BW), resulting in a decrease of BDNF overflow, responsible for reduced pain sensitivity and hyperalgesia in the adult rats.

Neuroinflammation also leads to excess BDNF production in the CNS, contributing to sensitivity in rodents.¹⁸ Exposure to psychostimulants affects cytokine production and their expression in the CNS and PNS.⁴⁵ Tobacco exposure causes a reduction of the anti-inflammatory cytokine IL-6.²⁰ In accordance with these earlier studies, the present study reveals downregulation of IL-6 expression during cigarette smoke and nicotine exposure (9.0 mg/kg BW) of 12 weeks, while BDNF expression becomes increased in both the 4- and 12-week groups as compared to the healthy unexposed controls (**Fig. 2E**). This complex and time-dependent effect of nicotine on these target molecules, especially IL-6 expression, suggests that short-term exposure to nicotine (4 weeks) may not be sufficient to trigger the sustained cellular changes required for significant alteration of the IL-6 production.⁴⁶ However, single post-exposure treatment of mecamylamine (3.0 mg/kg BW) exhibits appropriate production of IL-6, which may be due to the persistent exposure to cigarette smoke and nicotine (9.0 mg/kg BW) upto 12 weeks (**Fig. 2E-F**). Simultaneously, with increasing IL-6 production, there is a downregulation of BDNF-Trk- β signalling, resulting in a lowering of inflammation and pain sensitivity.^{47,48} This result is corroborated the synergistic effect of BDNF and IL-6 that opined nicotine itself induces inflammatory response along with the alteration of BDNF expression in the cerebral cortex microenvironment.^{19, 47,49}

Apart from IL-6, nicotine inhibits the release of other cytokines and inflammatory mediators like IL-1, IL-17, and Interferon-alpha (IFN- α), which are further suppressed by the free radicals such as myeloperoxidase, nitric oxide, and iNOS, as well as other immune-related responses, resulting in inhibition of the NF-B/Stat-3 signalling pathway and related factor, miR-21,^{24,25} leading to the development of neuroinflammation. iNOS alone is a proven key marker for oxidative stress that causes inflammation due to excessive production of nitric oxides.⁵⁰ This

evidence further supports our present notion that high expression of iNOS upon cigarette smoking and nicotine exposure (9.0 mg/kg BW) as compared to the healthy controls, or single post-exposure treatment of nAChR blocker, mecamylamine (3.0 mg/kg BW), irrespective of the duration of exposure (**Fig. 3A**).⁵¹ This observation may imply the disbalance of pro- and anti-inflammatory cytokines in the cellular microenvironment by free radical formation, which is manifested by increased pain perception during nicotine exposure. Moreover, chronic stress stimulates pro-inflammatory microglia, prompting cells to release cytokines and pro-inflammatory chemicals and recruit peripheral immune cells into the brain, resulting in the inflammatory milieu typical of brain diseases.¹⁸ This increased expression seems to be reduced by the nAChR blocker mecamylamine (3.0 mg/kg BW) in both the 4- and 12-week studies (**Fig. 3A-B**). In contrast, Nrf-2, as a master regulator of antioxidant defence mechanisms, directly influences redox homeostasis by regulating antioxidant enzymes or increasing the expression of antioxidant proteins.⁵² In the present precipitated withdrawal study, single post-exposure dose of mecamylamine (3.0 mg/kg BW) shows the anti-inflammatory property as Nrf2 expression is increased upon treatment as compared with the cigarette smoking and nicotine-exposed (9.0 mg/kg BW) groups of 4 and 12 weeks (**Fig. 3A-B**). High Nrf2 expression due to mecamylamine is thus responsible for the downregulation of iNOS in the cerebral cortex of the adult rats, which, in turn, activates IL-6 and reduces oxidative stress as well as neuroinflammation, thereby lowering pain sensitivity.⁵³

Conclusion

Overall, the present study is the first-ever study to emphasize the crucial role of BDNF-Trk- β and IL-6 coexpressions in cigarette smoking- and nicotine-induced hyperalgesia and neuropathic pain, mediated through disbalance of redox homeostasis and inflammation in the cerebral cortex (**Fig. 4**).^{19,54} The administration of mecamylamine at only a 3.0 mg/kg BW dose exacerbates pain sensitivity and ameliorates pain tolerance, thereby successfully mitigating the

negative effects of passive cigarette smoking and nicotine exposure at a higher dose of 9.0 mg/kg BW. Mecamylamine changes cytokine dynamics by upregulating IL-6 expressions, restoring redox balance through iNOS/Nrf2 levels, and downregulating BDNF-Trk- β expressions in the cerebral cortex neural circuits implicated for pain and inflammation (**Fig. 4**). Thus, the BDNF-Trk- β /IL-6 cascade may have a therapeutic potential for targeting the reduction of neuropathic pain, alleviating hyperalgesia and preventing nicotine relapse, so far as the smoking cessation and nicotine withdrawal are concerned. Further, the present research lays the groundwork for creating a focused non-opioid intervention strategy addressing both the aspects of nicotine addiction and pain management in order to exacerbate smoking cessation, or quitting success rates.

Supplementary Materials:

Submitted as a separate file.

Authors Contribution:

Vishakha Shrimali^{1,2}, Behavioral study, Laboratory investigation, Statistical analysis, Literature survey; Deepsi Rathore^{1,2}, Statistical analysis, Literature survey, Writing – original draft, Drew illustration; Vishakha Shrimali^{1,2} and Deepsi Rathore^{1,2}, Equally contributed as first author. Aanchal Joshi¹, Animal dosing, Literature survey. Nibedita Naha¹, Conceptualization, Visualization, Supervision, Fund Acquisition, Project administration, Resources, Animal dosing, Writing- original draft, Writing - review and editing. All authors read and approved the final manuscript.

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Data Availability Statement:

The original datasets were generated during the course of the project conducted at the ICMR-National Institute of Occupational Health (Ministry of Health and Family Welfare, Department of Health Research, Government of India). These datasets can be made available to readers upon reasonable request to the corresponding author, provided that the request includes a suitable rationale and complies with the journal's copyright policy.

Declaration of Interest:

None declared.

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Table 1. Study design and distribution of adult rats in different groups of the precipitated withdrawal study upon cigarette smoking and nicotine exposure for a period of 4 and 12 weeks. Parenthesis mentioned number of rats in each group. Details are mentioned in the *Materials and Methods* section.

Display Figure Legends

Figure 1. Effects of nAChR blocker, mecamylamine on hyperalgesia of the adult rats in tail-flick test upon cigarette smoking (A-B) and nicotine exposure (C-D) for a period of 4 and 12 weeks. Crt, Healthy unexposed controls. CS, Passive cigarette smoking through whole-body inhalation chamber, 2 cigarettes, 1 hr/day. N6-N9, Oral nicotine exposure through gavage, 6.0 and 9.0 mg/kg BW/day. Me1.5-Me3.0, Mecamylamine through i.p., 1.5 and 3.0 mg/kg BW, post-exposure. Details are mentioned in the *Materials and Methods* section. Results represent the mean $\pm$ SEM of the three independent experiments. n=5rats/group.

Figure 2. Effects of nAChR blocker, mecamylamine on nAChR, BDNF, Trk- β and IL-6 expressions of the adult rat's cerebral cortex by IHC upon cigarette smoking (A-F) and nicotine exposure (A-F) for a period of 4 and 12weeks. Crt, Healthy unexposed controls. CS, Passive cigarette smoking through whole-body inhalation chamber, 2 cigarettes, 1 hr/day. N6-N9, Oral nicotine exposure through gavage, 6.0 and 9.0 mg/kg BW/day. Me1.5-Me3.0, Mecamylamine through i.p., 1.5 and 3.0 mg/kg BW, post-exposure. Details are mentioned in the *Materials and Methods* section. Representative photomicrographs show colocalization of nAChR/BDNF (A: brown/black), Trk- β /BDNF (C: brown/black) and BDNF/IL-6 (E: brown/black) proteins in the cerebral cortex (x200). Results represent the mean $\pm$ SEM of the colocalization of nAChR/BDNF (B), Trk- β /BDNF (D) and BDNF/IL-6 (F) obtained from 10 randomly selected areas (1.5 $\times$ 1.5 cm each) of the cerebral cortex. n=5 rats/group.

Figure 3. Effects of nAChR blocker, mecamylamine on iNOS and Nrf2 expressions of the adult rat's cerebral cortex by IHC upon cigarette smoking (A-B) and nicotine exposure (A-B) for a period of 4 and 12weeks. Crt, Healthy unexposed controls. CS, Passive cigarette smoking through whole-body inhalation chamber, 2 cigarettes, 1 hr/day. N6-N9, Oral nicotine exposure through gavage, 6.0 and 9.0 mg/kg BW/day. Me1.5-Me3.0, Mecamylamine through i.p., 1.5

and 3.0 mg/kg BW, post-exposure. Details are mentioned in the *Materials and Methods* section. Representative photomicrographs show colocalization of iNOS/Nrf2 (A: brown/black) proteins in the cerebral cortex (x200). Results represent the mean $\pm$ SEM of the colocalization of iNOS/Nrf2 (B) obtained from 10 randomly selected areas (1.5 $\times$ 1.5 cm each) of the cerebral cortex. n=5 rats/group.

Figure 4. Role of nAChR blocker, mecamylamine hydrochloride on hyperalgesia and neuropathic pain management through BDNF-Trk- β /IL-6 cascade in the adult rats of cigarette smoking- and nicotine exposure for a period of 4 and 12 weeks, where nicotine doses closely mimic the real-life human smoking situation. BDNF-Trk- β /IL-6 cascade in cerebral cortex could be a non-opioid therapeutic target to improve smoking cessation/nicotine withdrawal effects. Details are mentioned in the *Materials and Methods* section.

Legends of Supplement Data

Figure S1. Flow diagram of study design and animal grouping for the precipitated withdrawal study upon cigarette smoking and nicotine exposure for a period of 4 and 12 weeks. Details are mentioned in the *Materials and Methods* section. n=5 rats/group.

Figure S2. Representative photomicrographs of negative control by IHC obtained from the unexposed healthy control rats (A-B x200). Cerebral cortex of the control group treated with secondary antibody alone, is served as negative controls. Experiments run in parallel to other study groups during IHC as mentioned in Figure 2-3. Details are mentioned in the *Materials and Methods* section. n=5 rats/group.

Table S1. Mann-Whitney U test of hyperalgesia of the adult rats by tail-flick test upon cigarette smoking and nicotine exposure for a period of 4 and 12 weeks. CS, Passive cigarette smoking through whole-body inhalation chamber, 2 cigarettes, 1 hr/day. N6-N9, Oral nicotine exposure through gavage, 6.0 and 9.0 mg/kg BW/day. Me1.5-Me3.0, Mecamylamine through i.p., 1.5 and 3.0 mg/kg BW, post-exposure. Details are mentioned in the *Materials and Methods* section. n=5 rats/group.

Table S2. Mann-Whitney U test of colocalization of nAChR, BDNF, Trk- β and IL-6 proteins in the cerebral cortex of the adult rats by IHC upon cigarette smoking and nicotine exposure for a period of 4 and 12 weeks. CS, Passive cigarette smoking through whole-body inhalation chamber, 2 cigarettes, 1 hr/day. N6-N9, Oral nicotine exposure through gavage, 6.0 and 9.0 mg/kg BW/day. Me1.5-Me3.0, Mecamylamine through i.p., 1.5 and 3.0 mg/kg BW, post-exposure. Details are mentioned in the *Materials and Methods* section. n=5 rats/group.

Table S3. Mann-Whitney U test of colocalization of iNOS and Nrf2 proteins in the cerebral cortex of the adult rats by IHC upon cigarette smoking and nicotine exposure for a period of 4 and 12 weeks. CS, Passive cigarette smoking through whole-body inhalation chamber, 2 cigarettes, 1 hr/day. N6-N9, Oral nicotine exposure through gavage, 6.0 and 9.0 mg/kg BW/day. Me1.5-Me3.0, Mecamylamine through i.p., 1.5 and 3.0 mg/kg BW, post-exposure. Details are mentioned in the *Materials and Methods* section. n=5 rats/group.

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Table 1. Outline of the study design and group distribution of adult rats in the precipitated withdrawal study, showing allocation across control, cigarette smoke/nicotine-exposure, and mecamylamine-treated groups for 4- and 12 weeks.

Figure 1. Graphs labelled A to D illustrate the effect of nAChR blocker mecamylamine on hyperalgesia using the tail-flick test in adult rat following 4- and 12 week exposure to cigarette smoke/nicotine and treated group (n=5 rats/group).

Figure 2. Representative IHC images and graphs with the labels from A-F. Adult rats' cerebral cortex photomicrographs of different protein colocalization (brown/black staining) following 4- and 12week exposure to cigarette smoke/nicotine, with or without mecamylamine treatment, are displayed under A, C, E label. B shows nAChR/BDNF, D Trk- β /BDNF and F shows BDNF/IL-6 quantification graphs that compares the expression levels across control, cigarette smoke/nicotine-exposed, and mecamylamine-treated groups, from 10 different regions of the cerebral cortex (n=5 rats/group).

Figure 3. Representative IHC images and graphs with the labels A and B. Adult rats' cerebral cortex photomicrographs of iNOS and Nrf-2 colocalization (brown/black staining) following 4- and 12 week exposure to cigarette smoke/nicotine, with or without mecamylamine treatment, are displayed under A label. B label shows quantification graphs that compare the expression levels of control, cigarette smoke/nicotine-exposed, and mecamylamine-treated groups, from 10 different regions of the cerebral cortex (n=5 rats/group).

Figure 4. Graphical illustration representing the role of nAChR antagonist, mecamylamine hydrochloride, in modulating hyperalgesia and neuropathic pain through the BDNF-Trk- β /IL-6 pathway in adult rats exposed to cigarette smoke/nicotine for 4 and 12 weeks mimicking real-life situation; underscoring its potential as a non-opioid therapeutic target in the cerebral cortex.

Alt Text for Supplement Data

Figure S1. Flow diagram representing the distribution of animals for the precipitated withdrawal study upon cigarette smoke/nicotine-exposure for a period of 4- and 12 weeks.

Figure S2. Photomicrographs of negative control by IHC obtained from the healthy rats unexposed control's cerebral cortex utilizing only secondary antibodies.

Table S1. Mann-Whitney U test results demonstrating hyperalgesia in adult rats treated with mecamlamine following 4- and 12 week exposure to cigarette smoke/nicotine and treatment with mecamlamine (n=5 rats/group).

Table S2. Mann-Whitney Results of the U test for the colocalization of nAChR, BDNF, Trk- β , and IL-6 protein expression in the cerebral cortex of adult rats following exposure to cigarette smoke/ nicotine along with mecamlamine treatment for 4- and 12 weeks (n=5 rats/group).

Table S3. Mann-Whitney Results of the U test for the colocalization of iNOS and Nrf-2 protein expression in the cerebral cortex of adult rats following exposure to cigarette smoke/ nicotine along with mecamlamine treatment for 4- and 12 weeks (n=5 rats/group).

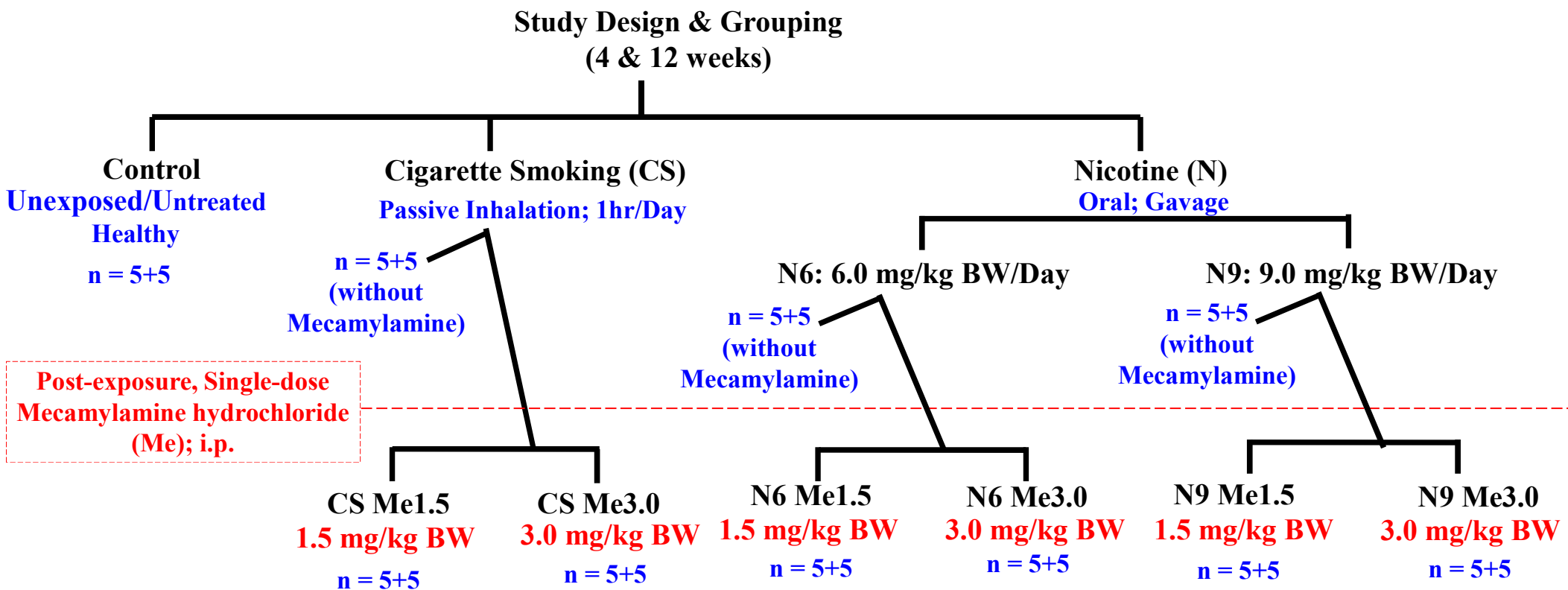


Fig. 1.

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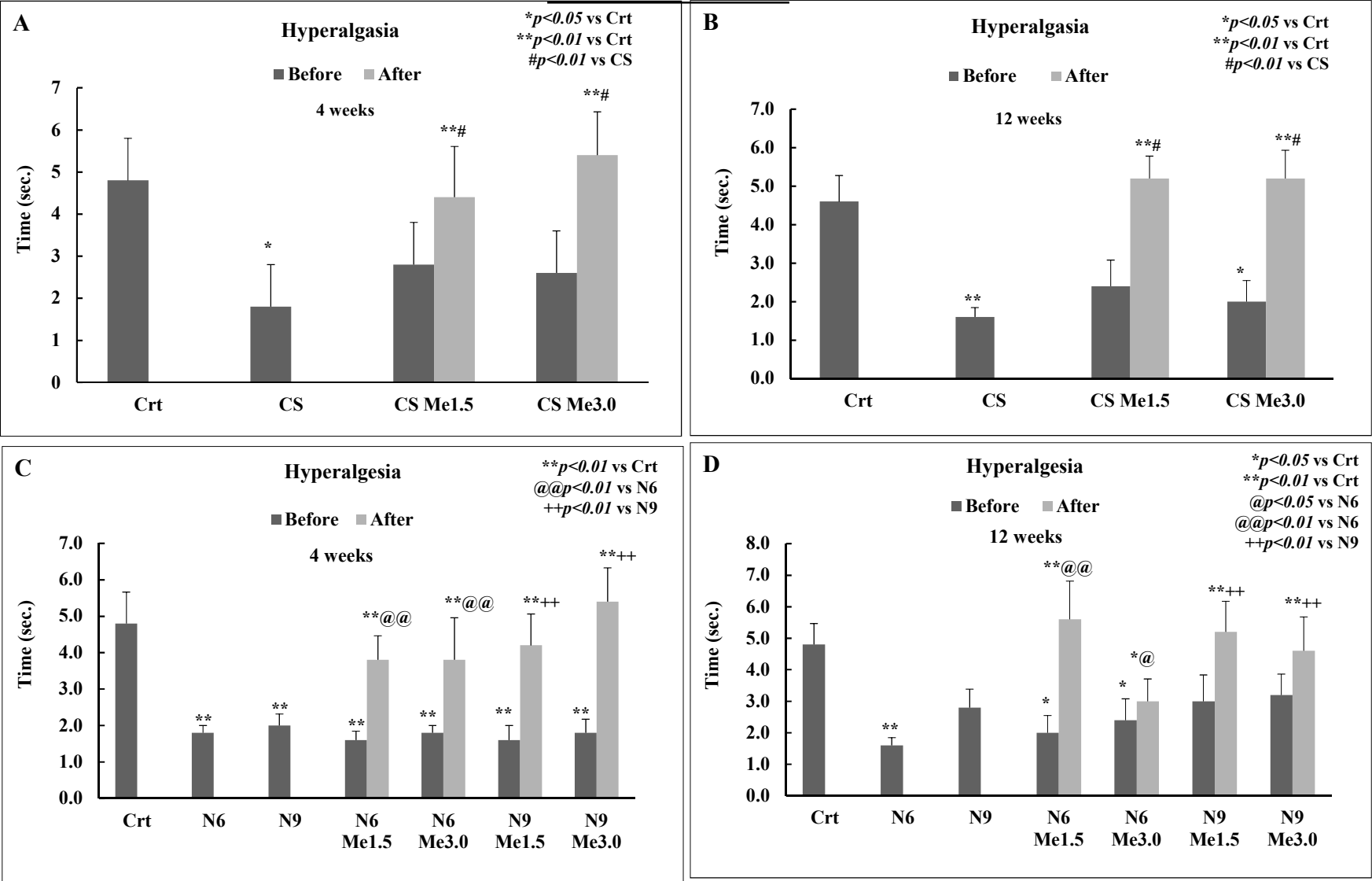


Fig. 2.

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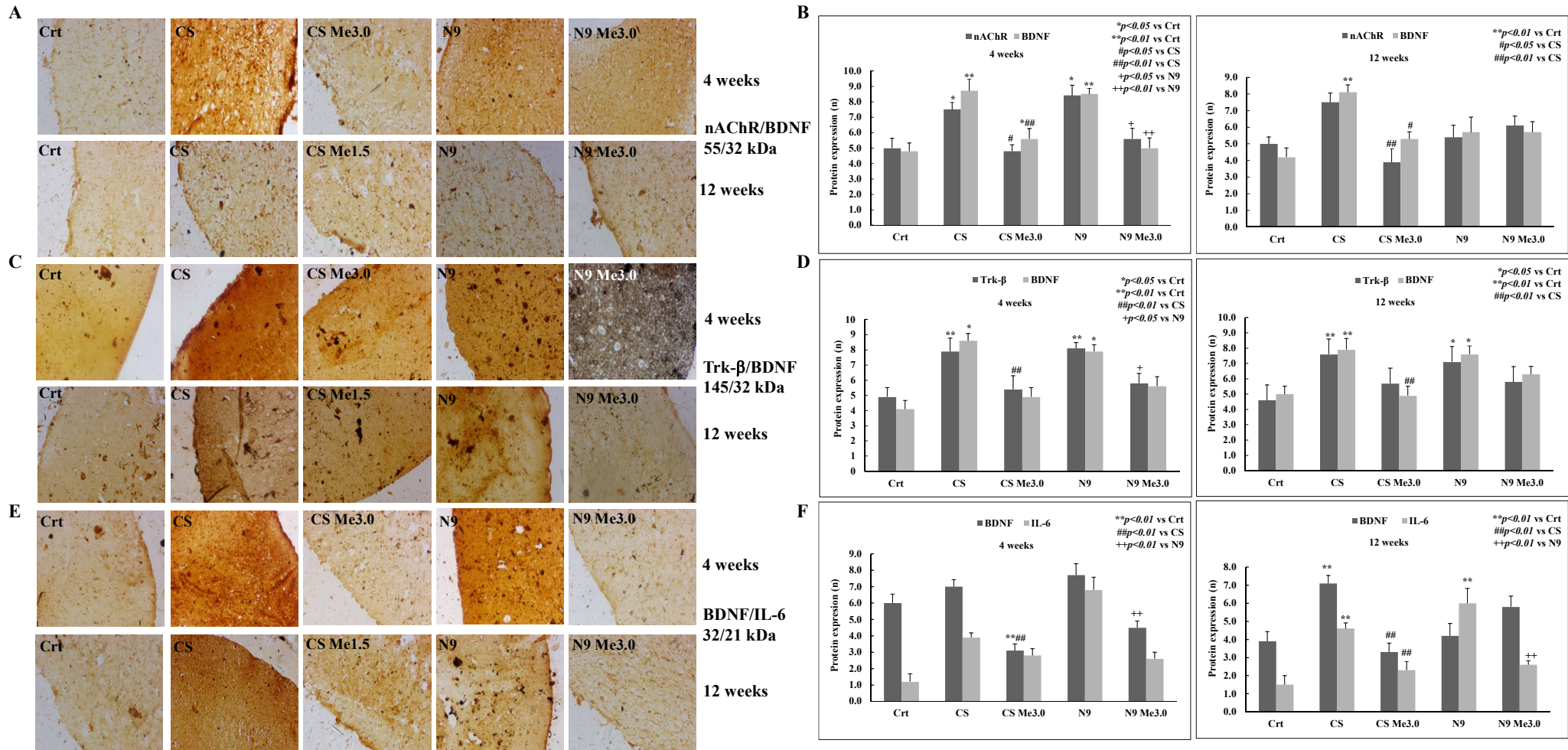


Fig. 3.

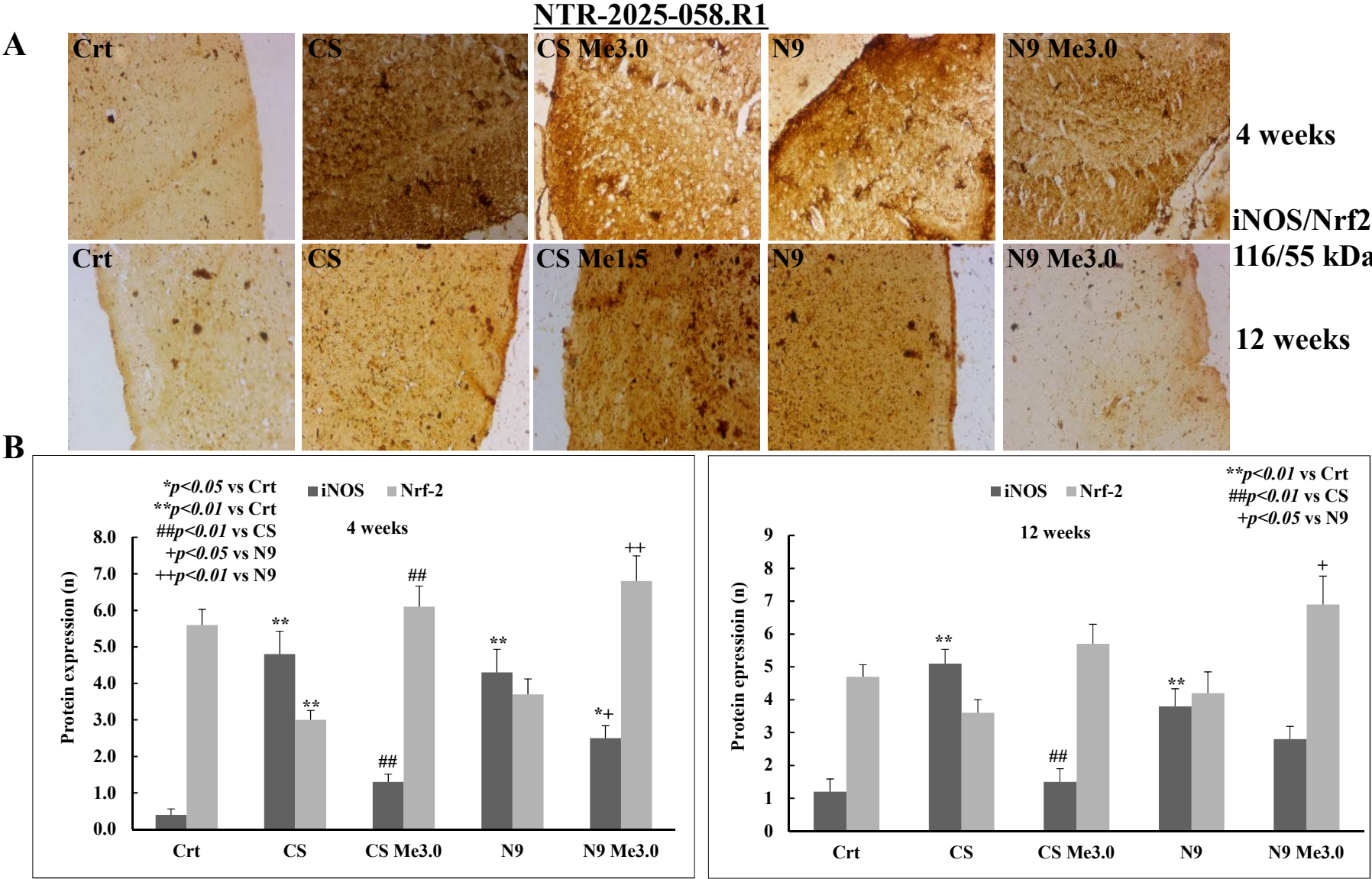


Fig. 4 Editable format

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